

RP3 quotient input: $n=5$ $p=2$ $q=1$

$X=0,1,2,3$

$O=2,3,5,0$

Kept generators in all Alexander gradings: 384

Building differential columns: 0/384

Computing matrix ranks...

Raw RP3 grid homology ranks over F_2 in all Alexander gradings

No V-factor cancellation has been performed.

$n=4$ $p=2$ $q=1$ components=1

$\text{Spin}^c = 0$

$F_2^{2,1} [M=-13/4, A=-17/4]$

$F_2^{2,4} [M=-9/4, A=-13/4]$

$F_2^{2,7} [M=-5/4, A=-9/4]$

$F_2^{2,7} [M=-1/4, A=-5/4]$

$F_2^{2,4} [M=3/4, A=-1/4]$

$F_2^{2,1} [M=7/4, A=3/4]$

rank = 24

$\text{Spin}^c = 1$

$F_2^{2,1} [M=-11/4, A=-15/4]$

$F_2^{2,4} [M=-7/4, A=-11/4]$

$F_2^{2,7} [M=-3/4, A=-7/4]$

$F_2^{2,7} [M=1/4, A=-3/4]$

$F_2^{2,4} [M=5/4, A=1/4]$

$F_2^{2,1} [M=9/4, A=5/4]$

rank = 24

Ranks by Alexander grading, summed over Spin^c and Maslov gradings:

$A=-17/4$ rank=1

$A=-15/4$ rank=1

$A=-13/4$ rank=4

$A=-11/4$ rank=4

$A=-9/4$ rank=7

$A=-7/4$ rank=7

$A=-5/4$ rank=7

$A=-3/4$ rank=7

$A=-1/4$ rank=4

$A=1/4$ rank=4

$A=3/4$ rank=1

$A=5/4$ rank=1

Total rank = 48